

Heart-failure v2 statistical handoff

Real-data demo · analysis, interpretation, and figure-reading guide

- Dataset:** UCI Heart Failure Clinical Records (n=299)
- Design:** Observational follow-up cohort; secondary analysis
- Primary outcome:** Time to death with censoring
- Review state:** needs-author-decision

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#	File	Claim
Figure 1	Fig1.png	Observed survival differs across prespecified ejection-fraction groups.
Figure 2	Fig2.png	EF and serum creatinine retain prognostic associations after prespecified adjustment.
Figure 3	Fig3.png	The two-feature model retains similar internal discrimination to the full baseline model.
Table 1	Table1.csv	The table describes, rather than balances, outcome groups.
Table 2	Table2.csv	The table reports effect size, uncertainty, and diagnostics together.

Figure 1 — Kaplan–Meier survival

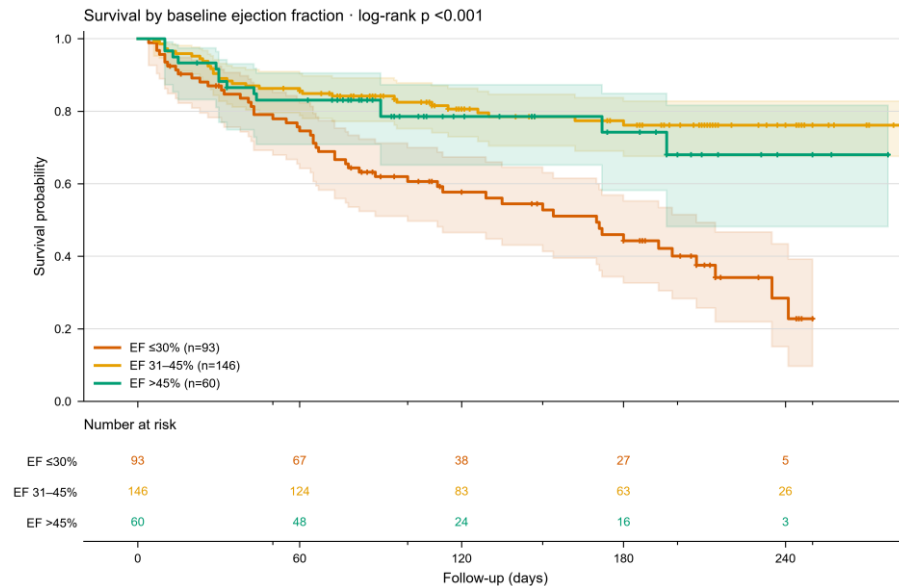
Claim: Observed survival differs across prespecified ejection-fraction groups.

Method rationale: Time-to-event outcome with censoring requires Kaplan–Meier estimation; the overall log-rank test compares the three curves.

Scientific result: Overall log-rank $p < 0.001$.

Interpretation: Lower baseline EF is associated with poorer observed survival.

Caption: Kaplan–Meier survival by prespecified baseline ejection-fraction group, with 95% confidence bands, censoring marks, and numbers at risk.



How to read

- Axes — follow-up days are on the horizontal axis; the vertical axis is the estimated probability of remaining alive.
- Colors — orange-red is EF ≤30%, gold is EF 31–45%, and green is EF >45%.
- Steps and lines — each downward step is an observed death; a horizontal segment means no death was observed between event times.
- Plus signs — these are censored patients, not deaths; follow-up ended without a recorded death at that time.
- Shaded bands — these are 95% confidence intervals around each survival estimate.
- Numbers at risk — the color-matched rows show how many patients remain under observation and event-free just before each time point.
- Reading order — compare when curves fall, how far they separate, how wide the bands are, and how many patients remain at risk.
- Late follow-up — treat tail separation cautiously when the risk rows contain only a few patients.

Common misreading: Curve separation is not causal evidence.

Cannot conclude: A treatment effect or a clinically optimal EF threshold.

Review status: final

Citation location: Results — survival analysis

Reproducibility: v2_demo/04_code/run_demo.py

Figure 2 — Adjusted Cox model

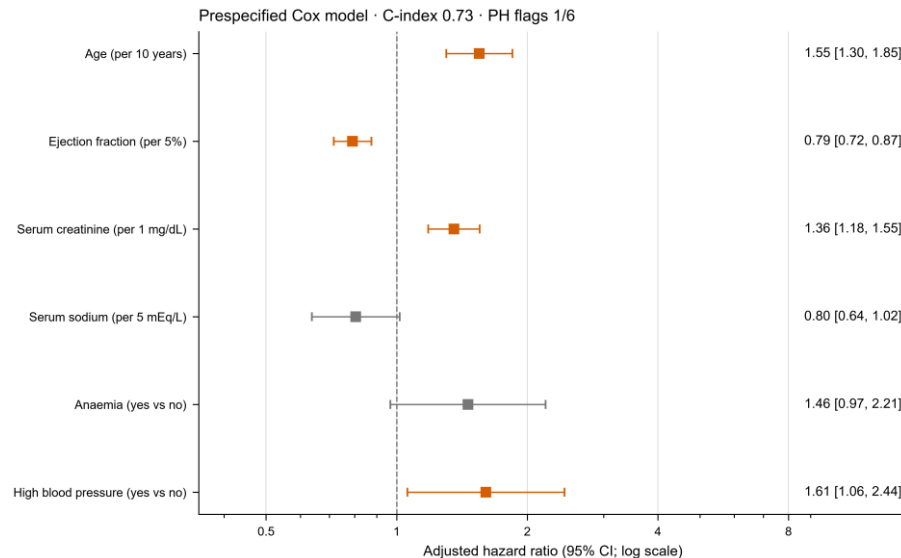
Claim: EF and serum creatinine retain prognostic associations after prespecified adjustment.

Method rationale: Cox regression estimates covariate-adjusted hazard ratios while retaining censoring information.

Scientific result: C-index 0.73; EF HR 0.79; creatinine HR 1.36.

Interpretation: Higher EF is associated with lower hazard; higher creatinine with higher hazard.

Caption: Prespecified multivariable Cox model. Continuous effects use clinically interpretable increments; horizontal lines are 95% confidence intervals.



How to read

- Rows — each row is one covariate in the same adjusted Cox model.
- Squares — the square marks the adjusted hazard ratio (HR); the horizontal whisker is its 95% confidence interval.
- Dashed line — HR=1 is the no-association reference. Intervals crossing 1 are compatible with no association at the 0.05 level.
- Direction — estimates left of 1 indicate lower hazard and estimates right of 1 higher hazard, conditional on the adjustment set.
- Scale — the horizontal axis is logarithmic, so equal distances represent equal multiplicative changes rather than equal absolute changes.
- Colors — orange denotes $p < 0.05$ in this figure and gray denotes $p \geq 0.05$; color does not establish causality or clinical importance.
- Units — EF is reported per +5 percentage points and creatinine per +1 mg/dL; other rows use the unit printed in the label.
- Diagnostics — read the C-index as model discrimination and inspect the PH flag before assuming an HR is constant over follow-up.

Common misreading: A hazard ratio is not an absolute risk ratio.

Cannot conclude: Causality, treatment benefit, or external validity.

Review status: needs-author-decision

Citation location: Results — adjusted associations

Reproducibility: v2_demo/04_code/run_demo.py

Figure 3 — Exploratory internal prediction

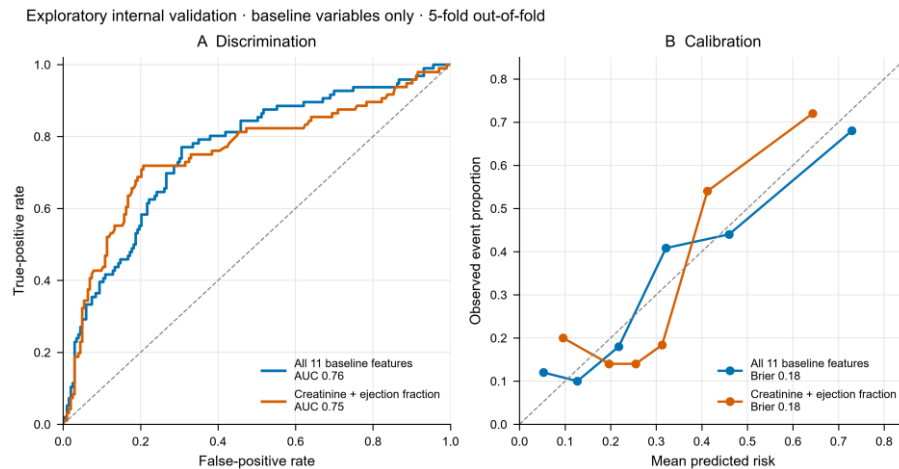
Claim: The two-feature model retains similar internal discrimination to the full baseline model.

Method rationale: Out-of-fold probabilities avoid evaluating each patient with a model trained on that patient; follow-up time is excluded.

Scientific result: Repeated-CV AUC: full 0.766 (0.061); two-feature 0.757 (0.063).

Interpretation: The compact model may be useful for hypothesis generation, not deployment.

Caption: Five-fold out-of-fold ROC and quantile-binned calibration for baseline logistic models. This secondary endpoint ignores unequal follow-up and is exploratory.



How to read

- Colors — blue is the full 11-feature model and orange is the two-feature creatinine + EF model in both panels.
- ROC panel — false-positive rate is on the horizontal axis and true-positive rate on the vertical axis; each curve position corresponds to a different threshold.
- ROC reference — the gray dashed diagonal is chance-level ranking; curves nearer the upper-left discriminate outcomes better.
- AUC — read the value in the legend as an overall ranking summary, not as accuracy at a selected threshold.
- Calibration points — each point is a quantile group of patients, not one patient; connecting lines are visual guides only.
- Calibration reference — the gray dashed identity line is perfect agreement. Points above it indicate underprediction and points below it overprediction for that group.
- Brier score — read the value in the legend as overall probability error; lower is better.
- Reading order — assess ROC discrimination and calibration separately, then compare both models without declaring equivalence from similar curves.

Common misreading: Similar AUC is not proof of model equivalence.

Cannot conclude: Transportability, net clinical benefit, or a safe decision threshold.

Review status: final

Citation location: Results — exploratory prediction

Reproducibility: v2_demo/04_code/run_demo.py

Table 1 — Cohort characteristics

Table 1. Baseline characteristics by observed outcome

Characteristic	Overall (n=299)	Censored (n=203)	Died (n=96)	SMD (Died – Censored)
Age, years	60.0 [51.0, 70.0]	60.0 [50.0, 65.0]	65.0 [55.0, 75.0]	0.54
Ejection fraction, %	38.0 [30.0, 45.0]	38.0 [35.0, 45.0]	30.0 [25.0, 38.0]	-0.58
Serum creatinine, mg/dL	1.1 [0.9, 1.4]	1.0 [0.9, 1.2]	1.3 [1.1, 1.9]	0.57
Serum sodium, mEq/L	137.0 [134.0, 140.0]	137.0 [135.5, 140.0]	135.5 [133.0, 138.2]	-0.41
Platelets, ×10 ³ /μL	262.0 [212.5, 303.5]	263.0 [219.5, 302.0]	258.5 [197.5, 311.0]	-0.1
Creatine phosphokinase, mcg/L	250.0 [116.5, 582.0]	245.0 [109.0, 582.0]	259.0 [128.8, 582.0]	0.12
Male sex	194 (64.9%)	132 (65.0%)	62 (64.6%)	-0.01
Anaemia	129 (43.1%)	83 (40.9%)	46 (47.9%)	0.14
Diabetes	125 (41.8%)	85 (41.9%)	40 (41.7%)	-0.0
High blood pressure	105 (35.1%)	66 (32.5%)	39 (40.6%)	0.17
Current smoking	96 (32.1%)	66 (32.5%)	30 (31.2%)	-0.03

Continuous: median [Q1, Q3]; binary: n (%). SMD direction is Died minus Censored.

Claim: The table describes, rather than balances, outcome groups.

Method rationale: Distribution summaries and standardized differences describe observed groups without p-value-driven screening.

Caption: Baseline characteristics of the public heart-failure cohort.

How to read

- Compare medians and IQRs for continuous rows.
- For binary rows, read numerator and percentage together.
- Use SMD magnitude and direction as a descriptive imbalance measure, not as a causal effect.

Common misreading: Outcome-group imbalance is not an independent prognostic effect.

Cannot conclude: Independent effects or causal risk factors.

Review status: final

Citation location: Methods and Results — cohort

Reproducibility: v2_demo/04_code/run_demo.py

Table 2 — Prespecified Cox model

Table 2. Multivariable Cox proportional-hazards model

Covariate	Hazard ratio	95% CI	p-value	PH-test p
Age, per 10 years	1.55	1.30–1.85	<0.001	0.83
Ejection fraction, per 5%	0.79	0.72–0.87	<0.001	0.025
Serum creatinine, per 1 mg/dL	1.36	1.18–1.55	<0.001	0.077
Serum sodium, per 5 mEq/L	0.8	0.64–1.02	0.068	0.142
Anaemia, yes vs no	1.46	0.97–2.21	0.071	0.921
High blood pressure, yes vs no	1.61	1.06–2.44	0.026	0.826

Covariate-level PH-test p uses rank-transformed time. A global diagnostic and graphical residual review remain pending; inspect any $p<0.05$ flag.

Claim: The table reports effect size, uncertainty, and diagnostics together.

Method rationale: Prespecified adjustment prevents p-value-driven variable selection.

Caption: Adjusted hazard ratios and proportional-hazards diagnostics.

How to read

- Read HR with its 95% CI, not p-value alone.
- Check the PH-test column before treating the HR as constant over follow-up.

Common misreading: Statistical adjustment does not remove unmeasured confounding.

Cannot conclude: A mechanistic or causal relationship.

Review status: needs-author-decision

Citation location: Results — adjusted associations

Reproducibility: v2_demo/04_code/run_demo.py